

<http://dx.doi.org/10.17703/JCCT.2025.11.3.545>

JCCT 2025–5–55

메타버스 체험 방식이 기억과 몰입감에 미치는 영향

Memory and Immersion in the Metaverse: An Experimental Study of PC, VR, and Viewing Modalities

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요약 본 연구는 메타버스 체험 방식이 사용자 기억력과 몰입도에 미치는 영향을 분석하고, 이를 바탕으로 교육 및 훈련 분야에 실질적인 시사점을 제시하는 데 목적이 있다. 총 122명의 참가자가 연구에 참여하였으며, 이들은 세 가지 메타버스 체험 방식 중 하나에 무작위로 배정되었다: (1) PC를 통한 상호작용(PC 집단), (2) VR 헤드셋을 이용한 상호작용(VR 집단), (3) 메타버스 환경 녹화 영상을 시청(Viewing 집단, 통제 집단). 기억력은 메타버스 체험 직후와 일주일 후 재인 과제를 통해 평가되었으며, 몰입도, 지각된 유용성, 자기 효능감 등의 사용자 경험 요소는 자기 보고식 설문으로 측정되었다. 연구 결과, 예상과 달리 Viewing 집단이 PC 집단 대비 통계적으로 유의미하게 높은 기억 정확도를 보였다. 이는 상호작용이 제한된 환경이 오히려 정보의 기억 유지에 긍정적 영향을 미칠 수 있음을 시사한다. 반면, 몰입도 측면에서는 PC 및 VR 집단이 Viewing 집단보다 유의하게 높은 수준을 나타냈으며, 이는 메타버스 환경과의 능동적 상호작용이 사용자의 몰입감 향상에 기여한다는 것을 입증하는 결과라 할 수 있다. 또한, 참가자들의 공간감 작업 기억 능력은 메타버스 환경에서의 기억 수행과 정적 상관관계를 나타내어, 개인의 인지적 역량이 몰입형 가상 환경에서의 학습과 기억 형성에 중요한 역할을 함을 입증하였다. 본 연구는 메타버스 기반 교육 및 훈련 프로그램 설계에 중요한 시사점을 제공하며, 효과적인 학습 환경 구축을 위해 사용자의 인지적 부하와 개인의 인지 능력을 고려한 맞춤형 접근이 필요함을 제안한다.

주요어: 메타버스, 체험 방식, 기억, 몰입도, 가상현실

Abstract This study examines the effects of different metaverse engagement modalities on memory and user experience, with implications for education and training. A total of 122 participants were randomly assigned to one of three groups: PC-based interaction (PC Group), VR-based interaction using head-mounted displays (VR Group), or video viewing of pre-recorded metaverse experiences (Viewing Group, control). Memory performance was assessed through immediate and delayed recognition tasks, while user experiences, such as perceived usability, were evaluated via self-reports. Results revealed that the Viewing Group demonstrated significantly higher memory accuracy than the PC Group, suggesting that environments with limited interactivity may positively influence information retention. Conversely, both PC and VR Groups reported significantly higher levels of immersion compared to the Viewing Group, indicating that active interaction with metaverse environments contributes to enhanced user engagement. No significant differences in immersion were observed between VR and PC Groups, suggesting that the immersive benefits of VR may vary depending on technological sophistication, task characteristics, and contextual factors. Additionally, visuospatial working memory positively correlated with memory performance, emphasizing the role of individual cognitive capacity in navigating immersive environments. These findings highlight the need for metaverse designs that balance cognitive accessibility with engaging interactivity.

Keywords: Metaverse, Modalities, Memory, Immersion, Virtual Reality

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Received: March 31, 2025 / Revised: April 22, 2025

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Accepted: May 2, 2025

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접수일: 2025년 3월 31일, 수정완료일: 2025년 4월 22일

게재확정일: 2025년 5월 2일

I . Introduction

The metaverse, a digital frontier that seamlessly blends virtual and augmented reality, is redefining human interaction, learning, and creativity by creating immersive environments where physical and virtual worlds converge [1]. Rooted in Neal Stephenson's 1992 novel *Snow Crash*, where it was envisioned as a vast interconnected digital landscape populated by user-controlled avatars [2], the metaverse has transformed from speculative fiction to a rapidly evolving reality. Enabled by advancements in virtual reality (VR) technologies, head-mounted displays (HMDs), motion tracking, and digital connectivity, it now serves as a platform for applications ranging from education to healthcare, entertainment, and commerce [3, 4, 5].

The concept of the metaverse revolves around the interplay between two pivotal constructs: immersion and presence. Immersion refers to the degree of sensory fidelity a virtual system provides, enabling users to feel enveloped in the virtual environment. In contrast, presence denotes the user's psychological experience of "being there" within the digital space [6]. These factors are essential for enhancing engagement and cognitive performance, forming the foundation of effective metaverse experiences. Such immersive qualities are particularly amplified by HMD systems, which provide a higher degree of sensory engagement than traditional desktop interfaces [7]. Recent findings suggest that higher levels of immersion can enhance episodic and spatial memory performance [8, 9, 10]. These immersive experiences likely facilitate deeper cognitive processing and reduce spatial ambiguity, thereby enabling richer encoding and more accurate retrieval of information. For instance, Krokos et al. (2019) demonstrated the effectiveness of HMD-enabled virtual memory palaces in enhancing recall accuracy, with participants achieving an 8.8% improvement compared to desktop conditions, illustrating the cognitive advantages of high immersion [11]. High-immersion settings further enhance these benefits; Schöne et al. (2019) found that autobiographical memory retrieval improved significantly in virtual environments, linking immersive learning to richer cognitive outcomes [12]. Additionally, VR platforms provide unique opportunities for collaborative learning, where users in remote locations interact in real-time within shared digital spaces [13, 14].

Despite its potential, the metaverse is not without challenges. The Cognitive Affective Model of Immersive Learning (CAMIL) proposed by Makransky and Petersen (2021) acknowledges that while presence and agency are key drivers of engagement and learning, excessive immersion can result in cognitive overload, undermining learning outcomes [15]. For example, Makransky et al. (2019) reported that although immersive simulations increased users' sense of presence, they simultaneously heightened cognitive load, which detracted from overall performance in educational tasks [7]. Richards and Taylor (2015) cautioned against over-reliance on 3D virtual environments, arguing that simpler 2D interfaces can sometimes offer superior outcomes by reducing unnecessary distractions [16].

This study seeks to address these complexities by systematically examining the cognitive effects of varying levels of immersion within metaverse environments. Specifically, it aims to compare different metaverse experience modalities—such as desktop-based versus HMD-enabled experiences—to evaluate how these variations influence memory retention and cognitive processing. The research questions guiding this investigation include:

What differences in cognitive processing occur among users based on their metaverse experience modality?

Does a high level of immersion consistently yield positive effects on cognitive processing and memory formation?

How do various user factors influence cognitive processing within metaverse environments?

By measuring user memory across diverse metaverse experiences while simultaneously assessing differences in immersion levels, this research aspires to elucidate the mechanisms underpinning cognitive processing in these environments. Ultimately, this study aims to contribute empirical insights that enhance our understanding of how to effectively leverage the metaverse and VR technologies for educational purposes while addressing challenges related to cognitive overload and accessibility.

II. Methods

1. Participants

A total of 166 participants were recruited for this study. Due to technical issues, such as difficulties in internet access or incomplete survey responses, data from 44 participants were excluded from the analysis. Consequently, the final dataset comprised 122 participants (Mage = 22.99, Nfemale = 70). All participants reported no history of brain injuries or diagnosed neurological disorders, nor any other medical conditions that could affect experimental outcomes. This study was approved by the Institutional Review Board of XX University (Approval No. 7001355-202203-HR-533).

2. Study Design

The study utilized a mixed design incorporating two primary factors. The first factor, a between-subjects variable, was the metaverse experience condition, which consisted of three experimental groups: (1) the PC access group, in which participants accessed the metaverse environment using personal computers (PC Group); (2) the VR access group, where participants engaged with the environment using VR HMDs (VR Group); and (3) the viewing-only group, in which participants viewed pre-recorded videos of the metaverse experience captured from the PC Group's activities, serving as the control condition (Viewing Group). This variable aimed to compare the effects of different metaverse experiences on participants' memory and user experiences.

The second factor, a within-subjects variable, was the memory condition. Participants completed recognition tasks at two-time points: immediately after the metaverse experience (immediate recognition) and one week later (delayed recognition). This design evaluated the retention of memory over time in a metaverse context.

The dependent variables were as follows: (1) recognition performance, assessed by participants' ability to accurately recall objects and content from the metaverse environment, and (2) metaverse user experience, which measured participants' immersion, perceived usefulness, and intention to use the metaverse. While the survey results compared differences across metaverse experience conditions, they were analyzed independently of the memory conditions.

1) Apparatus

For the PC Group, participants used their personal computers (desktops or laptops) in environments with sufficient specifications and stable internet connectivity. The VR Group utilized Meta Quest 2 HMDs with controllers, featuring a resolution of 1832×1920 per lens and a 120Hz refresh rate. The HMDs were used standalone without additional device connections. The Viewing Group used a 23.8-inch monitor (1920×1080 resolution, 60Hz refresh rate) to watch pre-recorded videos of the metaverse experiences. Participants in this group also used headphones to ensure clear audio during the viewing.

2) Experimental Stimuli

The metaverse environments were developed using Microsoft AltspaceVR tools and consisted of two settings. The first, a social interaction environment, included areas such as an auditorium for communication, a theater for video viewing, and recreational spaces with games like basketball and darts (Figure 1(a)). The second, an art appreciation environment, featured virtual galleries with paintings and sculptures, as well as simulated online and offline stores where participants could view or purchase items via embedded links (Figure 1(b)).

Participants explored both environments during a 20-minute guided session for each setting. Up to 10 participants accessed the environment simultaneously, accompanied by a developer to ensure smooth navigation. The order of environment exploration was counterbalanced to minimize order effects.

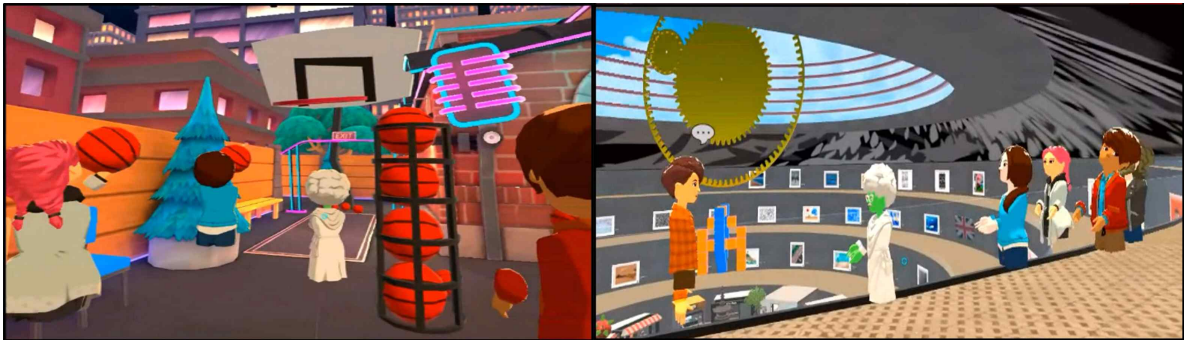


Figure 1(a) Example of metaverse social environment.

Figure 1(b) Examples of metaverse art environment.

3. Procedure

Participants began the experiment with a visit to the lab, where they provided informed consent and completed demographic surveys. To assess visuospatial working memory, participants performed the Picture Span Test (PST) [17], a task requiring memory of specific objects within a series of images (see details below). Upon completing the PST, participants were randomly assigned to one of the experimental conditions and proceeded to participate in the corresponding metaverse experience.

Participants assigned to the PC Group completed the experiment remotely. They were instructed to return home and access the metaverse platform using their personal computers at a designated time and convenient location. Within the platform, participants explored the metaverse environment alongside other participants. After engaging with the first metaverse environment, the experiment facilitator provided a link to an immersion survey, which participants were asked to complete. Participants were then instructed to take a short break before proceeding to the second metaverse environment. Upon completing the second environment, they filled out the same immersion survey, followed by an additional survey assessing memory performance and their overall metaverse experience.

Participants in the VR Group followed a slightly different protocol. After completing the PST, they were provided with details regarding the time and location of their VR experience and were sent home. At the designated time, participants convened at the specified location and used HMD devices to access the metaverse platform and explore the virtual environments. As with the PC Group, participants completed immersion surveys after exploring each environment and, immediately after the second environment, completed a memory task and the metaverse experience survey. For the VR Group, the facilitator also assessed participants for cybersickness following each environment by administering the Simulator Sickness Questionnaire (SSQ). Participants reporting

symptoms of cybersickness were given additional rest before continuing the experiment.

Lastly, participants assigned to the Viewing Group conducted their experiment in a controlled laboratory setting under the supervision of the experimenter. They watched 20-minute pre-recorded videos of the metaverse environments, captured directly from the PC Group's activities on their computer screens, in a quiet laboratory setting. After watching the first video, participants were asked if they experienced any fatigue. If they indicated no need for a break, they proceeded immediately to the next video. Upon completing each video, participants filled out an immersion survey evaluating their perception of the environment shown in the recording. Once all videos were viewed, participants completed a memory task followed by the metaverse experience survey, ensuring comprehensive feedback on their experience.

One week following the initial experimental session, all participants—regardless of group—were sent a survey link to complete a delayed memory task. Participants were compensated with 30,000 Korean won.

4. Measures

To account for potential confounding effects, participants' video streaming habits and VR experience frequency were assessed using single-item measures for each variable. Video streaming time was evaluated with the question: "How much time do you spend daily on online video streaming services (e.g., YouTube, Netflix)?" Responses were recorded on a 6-point Likert scale: 0 = None, 1 = Less than 2 hours, 2 = 2 - 3 hours, 3 = 3 - 4 hours, 4 = 4 - 5 hours, and 5 = More than 5 hours. VR experience frequency was assessed by first asking participants whether they had any prior VR experience. Those who responded affirmatively were subsequently asked to report the frequency of their experiences using a 6-point Likert scale: 1 = Fewer than 3 times, 2 = 3 - 5 times, 3 = 5 - 7 times, 4 = 7 - 10 times, and 5 = More than 10 times. Participants who indicated no prior VR experience were coded as 0 for the purpose of analysis.

1) Memory task

The memory task evaluated participants' ability to recognize objects introduced in the metaverse environments. Each question presented an image with a highlighted object, and participants were asked to indicate whether the object had been observed in the environment by responding "Yes" or "No" (Figure 2(a)). The task included 12 images from the first environment and 16 from the second. To assess false recognition, half of the images depicted objects that had been altered or were unrelated to the actual metaverse environments (Figure 2(b)). Correct responses were awarded 1 point, while incorrect responses received 0 points. The average recognition rate was then calculated and utilized for subsequent analyses. The immediate and delayed memory tasks used identical sets of questions.

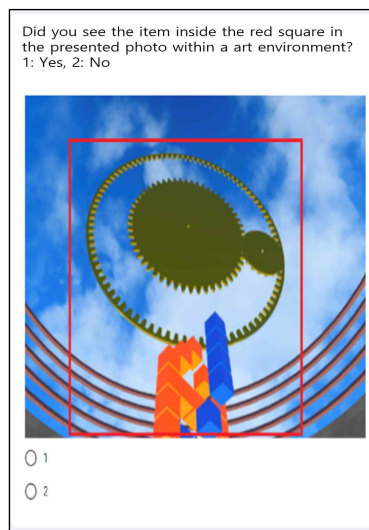


Figure 2(a). Example of a memory task employing an original object.

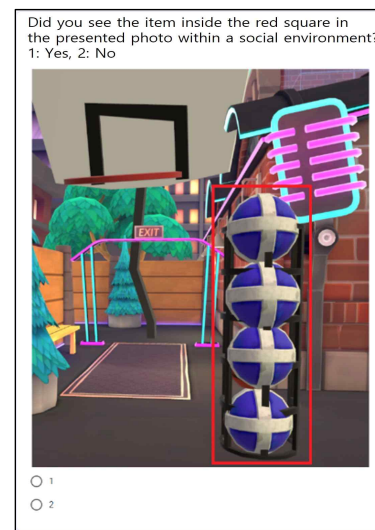


Figure 2(b). Example of a memory task employing an altered object.

2) Visual spatial working memory

The PST, developed by Tanabe and Osaka (2009) and utilized in this study with the authors' permission [17], is designed to assess visuospatial working memory. In this task, participants were required to remember specific objects displayed within a series of images while simultaneously counting aloud from 1 to 9, ensuring sustained cognitive engagement throughout the task (Figure 3.). The sequence of images presented varied in length, ranging from two to five images shown consecutively. Participants' visuospatial working memory capacity was evaluated based on their ability to accurately recall the objects, with scores ranging from 0 to 5.

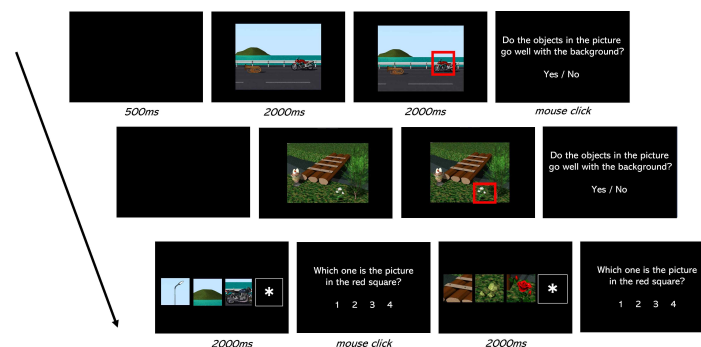


Figure 3. The procedure of the PST task, showing the steps involved in a 2-span sequence

3) Metaverse user experience

- **Immersion.** Immersion was assessed using three questions designed to measure the extent to which participants subjectively felt engaged in the metaverse environment. The specific items were as follows: "I felt like I was concentrating on the metaverse activity I just experienced," "I felt like I was immersed in the metaverse environment I just experienced," and "I felt like I was actively participating in the metaverse environment I just experienced." Responses were recorded on a 7-point Likert scale, ranging from 1 = not at all true to 7 = very true, and the average score across the three items was calculated. The internal consistency of the scale was high, with Cronbach's $\alpha = .923$.

- **Perceived Usability.** Perceived usefulness was measured using a scale that evaluates the extent to which a specific technology is perceived to improve quality of life compared to previous experiences [18, 19]. In this study, three items, adapted from the original five-item scale, were utilized based on modifications made in the previous study [20]. Each item assessed the degree to which metaverse usage was perceived as useful in improving quality of life, with the phrase “same as what I experienced today” added to the original items. Responses were collected on a 5-point Likert scale, ranging from 1 = not at all true to 5 = very true, and the sum of the three items was used for analysis. The reliability of the original scale was .911, while the Cronbach’s α for this study was .824.

- **Metaverse. Using Intention** To assess future intentions to use the metaverse, the Metaverse Use Intention Scale was utilized [21, 22]. The version of the scale employed in this study included items modified from Oh’s (2021) work [23]. Similar to the perceived usefulness items, the phrase “same as what I experienced today” was incorporated into each item. Participants responded on a 5-point Likert scale, ranging from 1 = not at all to 5 = very much, and the total score of the three items was used for analysis. The reliability of the scale in previous research was reported as .899, while the Cronbach’s α for this study was .904.

- **Perceived Enjoyment.** Perceived playfulness was assessed using a four-item scale modified by Oh (2021) [23] based on the items developed by Park and Byeon (2013) [24]. This scale evaluated the degree to which using the metaverse is perceived as enjoyable or fun. Each item was rated on a 5-point Likert scale, ranging from 1 = not at all to 5 = very much. The total score of the three items was calculated for analysis. In previous studies, the scale demonstrated a reliability of .940, while the Cronbach’s α for this study was .890.

- **Metaverse Self-Efficacy.** The Metaverse Self-Efficacy Scale was used to evaluate individuals’ confidence in acquiring skills, understanding terminology, and explaining functions required for effective metaverse use (e.g., “I feel confident in my ability to acquire skills related to the metaverse”) [25, 26]. The version of the scale employed in this study comprised three items adapted from the modifications made in Oh (2021) [23]. Participants rated their responses on a 5-point Likert scale, ranging from 1 = not at all to 5 = very much, and the total score of the items was calculated for analysis. The reliability of the scale in previous research was reported as .926, while the Cronbach’s α for this study was .826.

4) Simulator sickness questionnaire

The SSQ, developed by Kennedy et al. (1993) [27], is a widely used tool for assessing cybersickness and has also been adapted for measuring VR sickness in Korea [28]. The questionnaire comprises 16 items designed to evaluate three primary symptoms: oculomotor disturbances, disorientation, and nausea. Responses are recorded on a 4-point Likert scale, ranging from 0 = no symptoms to 3 = severe symptoms. In this study, the survey was conducted exclusively for the VR group, evaluating participants’ condition both before and after exposure to the virtual environment. The assessment aimed to determine whether participants’ symptoms fell within the typical range of simulator-induced motion sickness (19–55). In cases of severe symptoms, additional measures were implemented to provide relief and ensure participant well-being.

5. Data analysis

All statistical analyses were conducted using IBM SPSS Statistics 26.0. A one-way analysis of variance (ANOVA) was performed to examine group differences in demographic variables and visuospatial working memory scores across the experimental conditions. To assess differences in metaverse experience variables

among the experimental groups, a multivariate analysis of variance (MANOVA) was conducted. Recognition memory performance was analyzed using a mixed ANOVA, with the experimental condition (PC, VR, Viewing) as the between-subjects factor and recognition interval (immediate and delayed recognition) as the within-subjects factor. False recognition rates, defined as incorrectly identified items not previously seen, were also analyzed using a mixed ANOVA. Pearson product-moment correlation analyses were conducted to explore relationships among memory performance, visuospatial working memory, and metaverse experience variables. All statistical assumptions were checked, and Bonferroni corrections were applied to control for multiple comparisons where appropriate.

III. Results

The results of the one-way ANOVA conducted to examine group differences in participants' demographic characteristics are presented in Table 1. A total of 122 participants, divided into the PC Group ($n = 45$), VR Group ($n = 32$), and Viewing Group ($n = 45$), showed no statistically significant differences across key demographic variables. Specifically, there were no significant group differences in mean age, time spent on online video streaming, or the number of prior VR experiences. Additionally, visuospatial working memory scores did not significantly differ between groups ($F = 1.81$), confirming that the experimental groups were comparable in terms of working memory capacity.

Table 1. Background variables and visuospatial working memory across experimental conditions

	PC Group (SD)	VR Group (SD)	Viewing Group (SD)	F
N (female %)	45 (57.8)	32 (53.1)	45 (60)	-
Age	23.18 (2.61)	23.06 (2.55)	22.76 (2.60)	0.34
Video streaming time	2.53 (1.12)	2.28 (1.22)	2.31 (0.95)	0.24
VR experience	1.38 (0.49)	1.22 (0.42)	1.44 (0.50)	2.13
PST	1.84 (0.99)	2.16 (1.00)	2.24 (1.10)	1.81

The results of the mixed ANOVA conducted to examine differences in recognition performance across experimental conditions are presented in Table 2. While the main effect of experimental condition was not statistically significant at the conventional threshold ($F = 3.07$, $p = .05$, partial $\eta^2 = .049$), it approached significance, warranting further exploration. A Bonferroni post hoc analysis revealed that the overall recognition rate for the Viewing Group ($M = 0.70$) was significantly higher than that for the PC Group ($M = 0.65$, $p < .05$). However, no statistically significant differences in recognition rates were observed between the VR Group and the other groups. The main effect of time on recognition performance was statistically significant, with immediate recognition ($M = 0.72$) being significantly higher than delayed recognition after one week ($M = 0.62$, $F = 134.75$, $p < .001$, partial $\eta^2 = .531$). This pattern remained consistent even after controlling for the number of prior VR experiences as a covariate, confirming that differences in recognition task performance were not influenced by the frequency of prior VR use.

Table 2. Mean and standard error of immediate and delayed recognition memory by experimental conditions

	N	Immediate recognition		Delayed recognition	
		M	SE	M	SE
PC Group	45	0.70	0.02	0.60	0.02
VR Group	32	0.72	0.02	0.61	0.02
Viewing Group	45	0.75	0.02	0.65	0.02

False recognition, defined as the proportion of incorrectly identified items that were not previously seen, was also analyzed using a mixed ANOVA (results not shown in Table 2). The analysis revealed a significant main effect of time, with false recognition rates significantly lower for immediate recognition ($M = 0.32$) compared to delayed recognition ($M = 0.50$, $F = 173.44$, $p < .001$, partial $\eta^2 = .593$). These findings indicate that recognition accuracy declined over time, accompanied by an increase in false recognitions, highlighting temporal dynamics in recognition performance.

Table 3 presents the results of the MANOVA conducted to examine differences in metaverse experience variables across experimental conditions. The analysis revealed a significant effect of experimental condition on immersion ($F = 9.17$, $p < .001$, partial $\eta^2 = .141$), while no significant differences were found for other metaverse experience variables, including perceived usability, metaverse using intention, perceived enjoyment, and metaverse self-efficacy. The immersion variable was measured following participants' experiences in two metaverse environments, and since no significant differences were observed between the two spaces, the mean of the two ratings was used for analysis. Post-hoc comparisons of immersion using Bonferroni corrections indicated that the VR Group ($M = 5.20$) and the PC Group ($M = 5.37$) both reported significantly higher levels of immersion compared to the Viewing Group ($M = 4.31$, $p < .001$). However, the difference in immersion between the VR Group and the PC Group was not statistically significant.

Table 3. Mean and standard error of immediate and delayed recognition memory by experimental conditions

	PC Group (SE)	VR Group (SE)	Viewing Group (SE)	F
Immersion	5.37 (0.20)	5.20 (0.22)	4.31 (0.18)	9.17***
Perceived Usability	11.55 (0.45)	11.81 (0.49)	10.76 (0.41)	1.58
Metaverse Using Intention	11.13 (0.50)	10.63 (0.55)	9.67 (0.46)	2.43
Perceived Enjoyment	16.50 (0.60)	16.38 (0.65)	14.82 (0.55)	2.68
Metaverse Self-Efficacy	10.63 (0.46)	10.41 (0.50)	9.91 (0.42)	0.70

*** $p < .001$

The results of the correlation analysis between the primary variables measured in this study are presented in Table 4. Immediate recognition and delayed recognition scores both exhibited statistically significant positive correlations with visuospatial working memory ($r = .253$ to $.345$, $ps < .05$), indicating that memory task performance is closely related to visuospatial working memory capacity. In contrast, no significant correlations were observed between recognition performance (both immediate and delayed) and metaverse experience variables, including immersion, perceived usability, usage intention, perceived enjoyment, and metaverse self-efficacy. This suggests that performance on recognition tasks is not influenced by subjective experiences within the metaverse environment but is more directly tied to individual cognitive capacities such as visuospatial working memory. Additionally, significant correlations were observed among the metaverse experience variables, except for the correlation between immersion and metaverse self-efficacy, which was not statistically significant. Notably, immersion demonstrated moderate to strong correlations with perceived usability ($r = .631$, $p < .001$), usage intention ($r = .640$, $p < .001$), and perceived enjoyment ($r = .621$, $p < .001$), underscoring the

interconnectedness of these experiential dimensions. Similarly, perceived usability, usage intention, and perceived enjoyment showed strong interrelationships ($r = .810$ to $.832$, $ps < .001$). Metaverse self-efficacy was positively correlated with perceived usability ($r = .210$, $p < .05$), usage intention ($r = .248$, $p < .05$), and perceived enjoyment ($r = .318$, $p < .01$).

Table 4. Correlations between memory performance, visuospatial working memory, and metaverse experience variables

	1	2	3	4	5	6	7	8
1	–							
2	.579***	–						
3	.253**	.345***	–					
4	–0.065	–0.043	–0.081	–				
5	0.002	0.038	–0.011	.631***	–			
6	–0.044	–0.054	–0.027	.640***	.822***	–		
7	–0.038	–0.05	–0.11	.621***	.810***	.832***	–	
8	–0.037	–0.008	0.004	0.097	.210*	.248**	.318***	–

Note. 1 = Immediate recognition, 2 = Delayed recognition, 3 = visuospatial working memory, 4 = Immersion, 5 = Perceived Usability, 6 = Usage Intention, 7 = Perceived Enjoyment, 8 = Metaverse Self-Efficacy.

* $p < .05$, ** $p < .01$, *** $p < .001$

IV. Discussion

The findings of this study enhance our understanding of the cognitive and experiential impacts of different metaverse engagement modalities, focusing on memory retention, cognitive load, and user immersion. These results align with the study's purpose of examining how PC-based interaction, VR-based interaction, and video viewing conditions influence recognition memory and subjective user experiences [1, 29].

Participants in the Viewing Group demonstrated significantly higher memory accuracy than the PC Group, suggesting that reduced cognitive demands in less interactive settings may improve performance [15, 30]. This finding aligns with previous study, which argue that low-interactivity modalities, such as video-based learning, reduce extraneous cognitive load, leading to cognitive advantages [31]. However, while this reduced load supports memory retention, it may limit engagement compared to interactive environments, highlighting a trade-off between immersion and cognitive manageability. Although cognitive load was not explicitly measured, the experimental conditions naturally varied in cognitive demands. According to the CAMIL framework, presenting a broader field of view or requiring learners to actively search for information can elevate extraneous cognitive load. In the present study, participants in the Viewing Group observed the content passively, likely resulting in relatively lower extraneous cognitive load. Meanwhile, the PC and VR Groups engaged in tasks that may have introduced additional germane load, fostering deeper cognitive processing but increasing overall demands. This aligns with cognitive load theory, which distinguishes between intrinsic, extraneous, and germane load and their impacts on learning and retention [7, 32]. No significant differences in memory performance were found between the VR Group and the PC Group, challenging assumptions that VR necessarily provides cognitive advantages in all contexts [12]. Zou et al. (2024) highlighted that task-specific design and alignment with learning objectives are crucial in optimizing VR's cognitive benefits. The lack of differences observed here may result from task design rather than the inherent capabilities of the modalities [33].

Participants' visuospatial working memory demonstrated a significant positive correlation with both immediate

and delayed memory task performance. This suggests that individuals with greater working memory capacity performed better in recognition tasks, consistent with the CAMIL, which posits that cognitive load in virtual environments impacts learning and memory [15]. In the current study, the memory task was specifically designed to assess participants' ability to recognize not only the presence of particular objects but also their spatial locations and contextual attributes within the immersive environment. Given that metaverse environments are inherently three-dimensional and necessitate active navigational exploration, successful task performance largely depended upon effective spatial encoding and retrieval processes, both of which are central to visuospatial working memory. These results emphasize the pivotal role of individual cognitive capacity in managing cognitive demands. Makransky et al. (2019) reported that individuals with higher working memory capacity can better handle the cognitive load of VR, leading to enhanced memory outcomes. In the present study, working memory capacity likely influenced participants' ability to process and retain information within the metaverse, underscoring the need to design virtual experiences that account for individual differences to avoid overloading working memory [7].

The findings on false recognition further enrich our understanding of memory performance in metaverse environments by highlighting the temporal dynamics of recognition accuracy. Specifically, the observed decline in memory accuracy and increase in false recognition over time underscores the susceptibility of memory to interference as retention intervals lengthen. This aligns with theories of reconstructive memory, which propose that reliance on less precise memory cues over time can lead to errors in distinguishing previously encountered items from unfamiliar ones [34, 35]. These findings suggest that while immersive metaverse experiences may facilitate initial encoding through enriched sensory input, irrelevant or extraneous elements could induce interference during retrieval. Furthermore, the absence of structured rehearsal or reinforcement strategies in the current study likely contributed to the observed decline in delayed memory performance. Thus, to enhance long-term retention, it is crucial to minimize memory interference by incorporating repeated exposure or providing clearer contextual cues. Designing virtual experiences that account for temporal degradation of memory could enhance the effectiveness of educational and training applications in the metaverse.

Consistent with previous research, this study confirmed that both VR and PC modalities provide heightened immersive experiences compared to passive viewing [6, 11]. However, contrary to prior expectations [7, 36], there were no significant differences in immersion ratings between the VR and PC Groups. This outcome may be attributed to the specific characteristics of the task and interaction methods employed in the study. Both VR and PC groups experienced guided navigation and engaged with identical content through controllers, conditions that may have diminished the immersive advantage typically associated with VR. This interpretation aligns with previous findings indicating that the benefits of VR-based interaction decrease when navigation and interaction elements are standardized across conditions [37, 38]. Moreover, user familiarity with technology likely influenced immersion levels. Given that most participants were university students accustomed to traditional PC interfaces, PC-based interactions may have been perceived as more intuitive. Consistent with this notion, Parong and Mayer (2021) reported that the novelty of VR can sometimes act as a distraction, impeding effective information processing, especially for users who have not yet acclimated to the technology [39]. This outcome reinforces Wang et al.'s (2022) assertion that immersion is a multifaceted construct influenced by user agency, task relevance, and environmental factors [31]. The enhanced performance in the Viewing Group, combined with comparable immersion ratings across VR and PC, supports Beck and Morgado's (2023) findings on the importance of interactivity and task simplicity in shaping cognitive outcomes [40].

These findings underscore the need to tailor metaverse designs to balance cognitive load and user immersion. The superior memory performance in the Viewing Group suggests that low-interactivity modalities may be particularly effective when high cognitive retention is required. In educational contexts, such formats may be well suited for delivering complex content or supporting learners familiar with traditional instructional methods. In contrast, environments emphasizing engagement, creativity, or collaboration may benefit from more interactive designs [31, 41, 42]. Given that responses to interactive elements can vary based on individual cognitive capacity, the observed correlation between working memory and memory performance underscores the need to tailor design complexity accordingly. Specifically, aligning interactive tasks with individual cognitive capabilities can prevent working memory overload and thus optimize outcomes. Incorporating user agency, task relevance, and narrative coherence can further enhance user experiences while minimizing unnecessary cognitive load [33, 43]. Such considerations are equally relevant in commercial metaverse applications, where adjusting interactivity to match user familiarity, cognitive profiles, and learning objectives can significantly improve both engagement and effectiveness.

Several limitations should be noted. First, the small sample size in the VR Group may have limited statistical power, obscuring nuanced differences. A post hoc power analysis conducted using effect sizes derived from the mixed ANOVA on memory performance in this study (calculated via G*Power) indicated that a minimum sample size of 28 participants per group was required. Although this study satisfied that criterion, the relatively smaller sample size of the VR group compared to the other groups may have limited the statistical sensitivity needed to detect subtle effects. Second, environmental control varied across groups, such as the Viewing Group's laboratory setting versus the VR and PC Groups' more variable contexts, which may have influenced outcomes. Third, reliance on recognition memory tasks limited the scope of cognitive assessment, not capturing dimensions like recall or spatial memory. Additionally, participants were not informed beforehand about the memory task to avoid inducing rehearsal strategies and altering natural engagement. However, this may have influenced how participants allocated attention during the metaverse activities, affecting memory performance. Finally, as López-Belmonte et al. (2023) highlight, user preferences and individual cognitive traits warrant further exploration [44].

Future research should address these limitations by ensuring standardized experimental conditions across groups to minimize variability. Expanding cognitive assessments to include recall, spatial memory, and executive function would provide a more comprehensive evaluation of cognitive impacts. Given the observed relationship between working memory and task performance, future studies should investigate how working memory interventions or scaffolds can alleviate cognitive load and enhance memory. Incorporating direct measures of cognitive load, such as the NASA-TLX, and examining its interaction with working memory could offer deeper insights into optimizing metaverse environments. Additionally, longitudinal studies are needed to assess the long-term effects of metaverse engagement on learning, memory, and user behavior. Finally, exploring individual differences such as prior VR experience and cognitive styles can refine insights into designing effective, accessible metaverse experiences.

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※ This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea (NRF) (No. NRF–2018S1A5B6069474).